AYON BISWAS

AI Engineer · Generative AI · RAG Systems · Agentic AI · LLM Integration
Gurugram, India | +91-8617607547 | ayonbiswas567@gmail.com
linkedin.com/in/ayonbiswas | github.com/ayonBiswas | AyonBiswas.dev/portfolio

PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience building production-grade Generative AI systems, specializing in Retrieval-Augmented Generation (RAG), Agentic AI workflows, and LLM-powered applications for enterprise clients including Lockheed Martin and Boeing. Proven ability to architect end-to-end AI pipelines — from custom data ingestion and hybrid search to fine-tuning, model evaluation, and cost-optimized deployment on Azure & AWS. M.Tech (AI) from NIT Rourkela.

TECHNICAL SKILLS

AI / ML: Large Language Models (LLMs), Generative AI, RAG, Prompt Engineering, Fine-tuning (LoRA, QLoRA), NLP, Computer Vision, Deep Learning, Transformer Architecture, CNNs, Transfer Learning, RLHF
Frameworks & Tools: PyTorch, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, LangSmith, OpenAI API, Groq, Ollama, Model Context Protocol (MCP), Scikit-learn
Agentic AI: Multi-Agent Orchestration, AI Agents, Tool Use, Agentic Workflows, smolagents, MCP Servers
Data & Vector DBs: Qdrant, FAISS, ChromaDB, Pinecone, PostgreSQL, SQL, SQLite FTS5, Semantic Search, BM25
Cloud & MLOps: Azure (AI Foundry, AKS, Functions, DevOps), AWS (EC2, Lambda, Bedrock), Docker, Kubernetes, Terraform, CI/CD, MLflow, LLMOps, Model Monitoring, Weights & Biases
Languages: Python (FastAPI, asyncio), C#, ASP.NET Core, SQL, JavaScript, Git/GitHub
Evaluation: Model Evaluation (Evals), Recall@k, Precision@k, MRR, NDCG, Guardrails, Responsible AI, Red-teaming, Drift Detection, Azure Monitor, Application Insights

PROFESSIONAL EXPERIENCE

AI Engineer — R&D | *Keysight Technologies* | *Gurugram* *Mar 2024 – Present*
Enterprise AI solutions for Lockheed Martin, Boeing, and Emirates | LINK

- **Architected and deployed an enterprise-grade RAG system over 500+ hardware documents,** reducing engineer manual search time by 70% and mitigating LLM hallucinations via verifiable source grounding.
- **Designed a Hybrid Search Architecture** combining dense semantic retrieval (Qdrant, HNSW/ANN) with sparse BM25 keyword matching (SQLite FTS5), improving retrieval precision by 35% over semantic-only baselines.
- **Implemented Reciprocal Rank Fusion (RRF)** with cross-encoder reranking (FlashRank), achieving Recall@10 of 92% and MRR of 0.87 on internal eval benchmarks.
- **Engineered a delta-ingestion pipeline** using SHA-256 content hashing, cutting re-ingestion time from minutes to seconds by only re-embedding modified chunks.
- **Reduced operational costs by packaging the full AI stack (Streamlit, onnxruntime, Qdrant)** into a 950 MB standalone portable executable, eliminating cloud SaaS dependency and saving $5,000+/year.

Software Engineer — R&D | *AMETEK Mocon (USA BU)* | *Bengaluru* *Oct 2023 – Mar 2024*

- Developed and maintained core features for an enterprise .NET application (C#, ASP.NET MVC), serving 200+ daily users.
- Optimized SQL Server queries and stored procedures, reducing application load time by 60%.

PROJECTS

Job Application Automation — MCP Server *[GitHub]*

- Built an Agentic AI system using the Model Context Protocol (MCP) to automate end-to-end job application workflows — including job discovery, resume tailoring, cover letter generation, and application submission.
- Designed multi-tool MCP server architecture with LLM-powered agents (LangGraph, OpenAI API) for autonomous decision-making, web scraping, and document generation; reduced manual application effort by 80%.
- Implemented structured output parsing, prompt engineering with guardrails, and CI/CD deployment via Docker.

Brain Tumor Detection & Classification — Deep Learning / Computer Vision *[GitHub]*

- Developed a CNN-based medical image classification pipeline in PyTorch to detect and classify brain tumors (glioma, meningioma, pituitary) from MRI scans, achieving 96%+ test accuracy.
- Applied transfer learning (ResNet-50), data augmentation, and hyperparameter tuning; built evaluation suite with confusion matrices, ROC-AUC, and Grad-CAM explainability visualizations.
- Deployed inference API using FastAPI and Docker; integrated model versioning with MLflow for experiment tracking.

EDUCATION

M.Tech — Signal & Image Processing | *NIT Rourkela* *2022 – 2024*
CGPA: 7.34 | Coursework: Machine Learning, Deep Learning, Computer Vision, Digital Signal Processing

B.Tech — Electronics & Communication Engg. | *Haldia Institute of Technology* *2018 – 2022*
CGPA: 8.75 | Qualified GATE ECE in Pre-final Year